

Convolutional Encoder-Decoder-Based Electric Field Intensity Solver for TGV Chip Arrays

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Abstract—This paper presents an electric field intensity solver based on a convolutional encoder-decoder, designed to overcome the meshing challenges and computational bottlenecks associated with Through-Glass Via (TGV) chip arrays. By establishing a direct mapping from geometric and frequency parameters to physical field distributions, the proposed method incorporates a feature-preserving mechanism that effectively mitigates the loss of fine boundary details inherent to conventional convolutional networks during downsampling. Experimental results demonstrate that the model achieves high-fidelity field predictions with minimal error, significantly accelerating large-scale parametric sweeps and optimization in 3D heterogeneous packaging design.

I. INTRODUCTION

Driven by the evolution of 3D heterogeneous packaging, TGVs have emerged as indispensable vertical interconnects for RF and microwave components, owing to their superior high-frequency transmission characteristics and low dielectric loss. However, in practical applications, high-density TGV arrays operate within complex electromagnetic environments where proximity effects and parasitic coupling severely impact signal integrity. Consequently, the accurate characterization of internal electric field distributions is critical for design optimization.

Currently, electromagnetic analysis of such structures relies primarily on full-wave numerical tools such as the Finite Element Method (FEM). While FEM offers high accuracy, the characteristic high aspect ratio of TGVs necessitates extremely dense mesh generation to capture geometric details. This results in a drastic increase in computational degrees of freedom and excessive memory consumption. The prohibitive simulation time per run renders conventional methods ill-suited for the extensive parametric sweeps and rapid iterations required in modern circuit design.

To address these efficiency bottlenecks, deep learning-based surrogate models have attracted significant attention. Nevertheless, conventional convolutional networks often suffer from the loss of critical geometric boundary information during downsampling operations, limiting their prediction accuracy for complex TGV structures. In response, this paper proposes an electric field intensity solver based on a spatial feature-preserving convolutional encoder-decoder. By incorporating a specialized feature-preserving mechanism, the proposed method constructs an end-to-end mapping that effectively retains spatial topological details. This approach achieves high-fidelity prediction of electric field distributions in milliseconds, provid-

ing a novel and efficient tool for electromagnetic compatibility design.

II. THEORY

A. Formula

The electromagnetic analysis of TGV chip arrays is fundamentally a boundary value problem for Maxwell's equations solved over a given geometric domain Ω and its boundary $\partial\Omega$. Considering the operating characteristics of microwave devices, this paper models the electric field distribution in the frequency domain.

Within the computational domain Ω , assuming an isotropic medium, the electric field intensity $\mathbf{E}$ obeys the vector wave equation. For regions containing non-zero source terms, the governing equation is expressed as:

$$\nabla \times \left(\frac{1}{\mu_r} (\nabla \times \mathbf{E}) \right) - k_0^2 \epsilon_{rc} \mathbf{E} = -jk_0 Z_0 \mathbf{J} \quad (1)$$

Where μ_r denotes the relative permeability. The complex relative permittivity is defined as $\epsilon_{rc} = \epsilon_r - j\sigma / (\omega\epsilon_0)$ characterizing the dispersion and loss properties of the medium; $k_0 = \omega\sqrt{\mu_0\epsilon_0}$ represents the free-space wavenumber, and Z_0 is the free-space wave impedance; $\mathbf{J}$ denotes the impressed current density source. This equation describes the propagation and scattering behavior of electromagnetic waves within the complex dielectric environment of the TGV array. To ensure the uniqueness of the solution, appropriate boundary conditions must be imposed on the computational domain boundary $\partial\Omega$. For the signal input and output ports of the TGV array, first-order absorbing boundary conditions or impedance boundary conditions are employed to simulate energy injection and matching. The boundary equation at the ports is defined as:

$$\mathbf{n} \times \left(\frac{1}{\mu_r} \nabla \times \mathbf{E} \right) = \frac{j\omega\mu_0}{Z_s} \mathbf{n} \times \mathbf{E} \times \mathbf{n} - 2 \frac{j\omega\mu_0}{Z_s} \mathbf{n} \times \mathbf{E}_0 \times \mathbf{n} \quad (2)$$

where $\mathbf{n}$ is the outward unit normal vector, Z_s is the port impedance, and $\mathbf{E}_0$ is the incident electric field vector. The first term on the right-hand side characterizes the reflected field and loss at the port, while the second term introduces the incident wave excitation as a source. For the metal interconnection layers and ground shielding layers within the TGV structure, the

surfaces are treated as Perfect Electric Conductors , satisfying the Dirichlet boundary condition where the tangential electric field vanishes:

$$\mathbf{n} \times \mathbf{E} = \mathbf{0} \quad (3)$$

B. Model

To preserve the critical boundary information of TGV arrays, the initial stage of the model avoids immediate downsampling. Instead, consecutive convolutional layers are employed to extract geometric features while maintaining the original spatial resolution. Subsequently, the network adopts a multi-level downsampling strategy, utilizing strided convolutions to progressively compress spatial dimensions while expanding channel depth. Through multiple abstraction stages, the input is encoded into a high-dimensional latent feature tensor. The decoder is tasked with reconstructing the physical field distribution from these abstract semantic features. This path utilizes transposed convolutions for upsampling, with strides strictly symmetric to the encoder to ensure the precise recovery of spatial dimensions. Finally, a terminal convolutional layer serves as a regression head, mapping the multi-channel feature maps to a single-channel continuous scalar field, yielding an electric field intensity prediction consistent with the input dimensions.

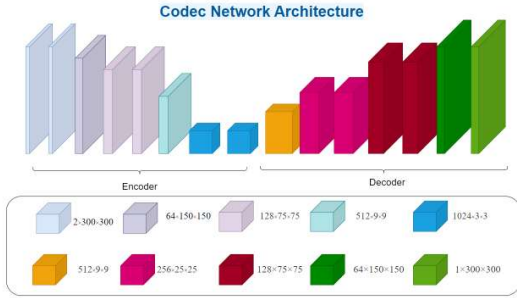


Figure 1. The proposed network architecture diagram.

III. RESULTS AND ANALYSIS

A. Data Generation and Training

Physically, the TGV array manifests as a 3D heterogeneous stack structure comprising the chip, glass interposer, and circuit substrate. To reduce input dimensionality while retaining critical physical features, we employ a geometric feature projection strategy. Although the structure involves multiple dielectric layers, the interposer and substrate constitute a relatively stable electromagnetic background. Therefore, our focus centers on the chip layer, which dominates the electric field distribution. By projecting the 3D structure along the Z-axis, the chip surface geometry is discretized into a geometric matrix. This matrix exclusively describes the metal pattern distribution of the chip layer, while the dielectric properties of the glass and substrate are implicitly embedded within the simulation constraints rather than serving as independent input channels. Subsequently, this geometric matrix is stacked with a frequency matrix of identical dimensions to form the 2D input tensor.

Based on the physical model of the TGV chip array (Fig. 2), chip dimensions and operating frequencies were selected as key variables and subjected to discrete stepping sweeps within

the ranges specified in Table I. For each parameter set, the chip surface geometry is mapped to a binary pixel matrix (assigned 1 where the chip is present, and 0 otherwise). The corresponding operating frequency value is expanded to fill a frequency matrix of the same size. These two matrices are concatenated along the channel dimension to construct the model's input tensor. A total of 4,961 samples were generated and partitioned into training, validation, and test sets in an 8:1:1 ratio.

Network weights were initialized using the Kaiming method, with biases set to zero. The model was trained using the Adam optimizer with a batch size of 64 for 128 epochs. Smooth L1 Loss was selected as the objective function, with Mean Absolute Error (MAE) introduced as the evaluation metric. Furthermore, an optimal model checkpointing strategy was implemented, where weight persistence occurs only when the validation MAE achieves a new historical low, ensuring optimal generalization performance. All experiments were implemented using the PyTorch framework on an NVIDIA RTX 4060 Ti GPU.

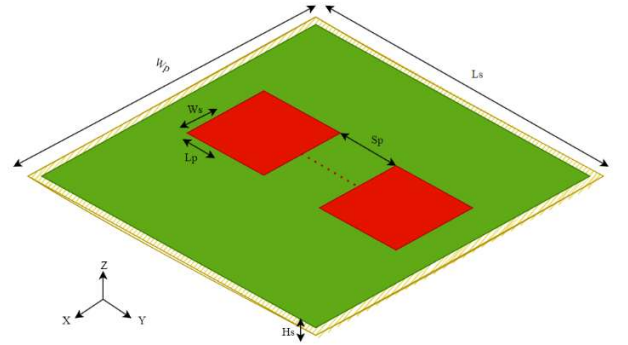


Figure 2. TGV Chip Array Model.

TABLE I. DATA SET GENERATION METHOD.

Name	Range	Interval	Quantity
L_p	7.5-10.5mm	0.3mm	11
W_s	4.5-6.5mm	0.2mm	11
f	10-10.8GHz	0.02GHz	41

B. Result

Fig. 3 illustrates the convergence trajectory of the loss function during training. The training and validation loss curves exhibit a rapid descent and maintain high consistency, indicating that the model achieves fast convergence with superior generalization capability, devoid of significant overfitting.

To evaluate the model's predictive performance on unseen data, Fig. 4 compares the ground truth electric field distributions with the model predictions from the test set. Visual inspection confirms that the predictions align closely with the ground truth, particularly in localizing peak field intensities and reconstructing edge gradients. The pixel-wise absolute error heatmap displays a global distribution of "cold" tones, representing low residual levels, with no observable singularities or structural biases.

Table II quantitatively compares the computational costs of the proposed encoder-decoder model against COMSOL. In contrast to the approximately 12 seconds required for a single full-wave simulation in COMSOL, the proposed model completes forward inference in just 66.7 ms, representing a 94.4% reduction in time costs. Furthermore, peak memory usage is limited to 171.5 MB, a substantial 94.9% reduction compared to the traditional FEM solver.

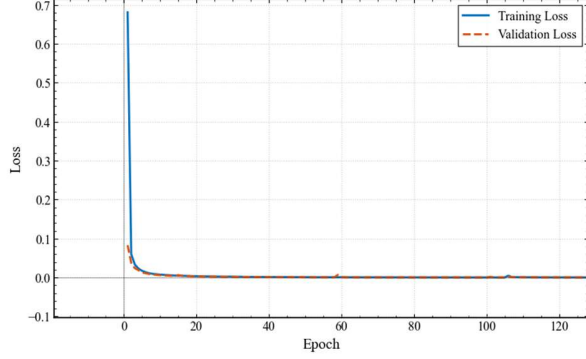


Figure 3. Loss Function Plot.

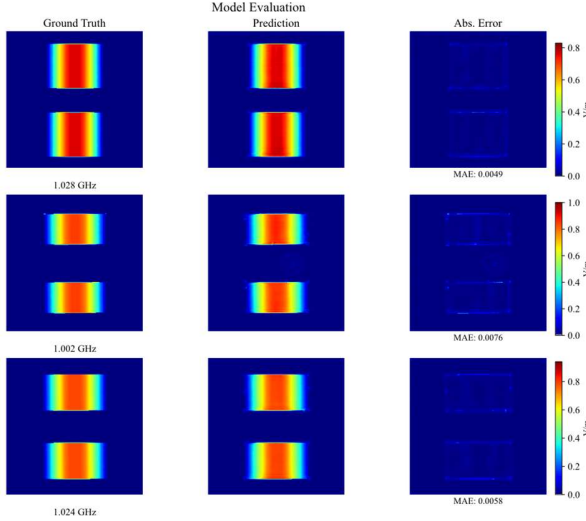


Figure 4. Model Evaluation Image.

TABLE II. COMPARISON WITH COMSOL COMPUTATIONAL SOFTWARE COSTS

	Time	Memory	Save time	Save Memory
COMSOL	12s	3379MB		
Proposed Net	66.7ms	171.5MB	94.4%	94.9%

IV. CONCLUSIONS

This paper presents a spatial feature-preserving convolutional encoder-decoder tailored for the electromagnetic analysis of TGV chip arrays. By establishing a direct mapping from geometric topology to physical fields, the proposed method effectively overcomes the efficiency bottlenecks of traditional numerical simulations and mitigates the loss of fine geometric details inherent in conventional deep networks. Experimental validation confirms that the solver achieves high-fidelity predictions with minimal residual levels.

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